

# STATS 201A Lecture 4

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$$H_0 : \tau_B - \tau_A = 0$$

$$H_A : \tau_B - \tau_A > 0$$

$$\sigma_1^2 = \sigma_2^2 = \sigma^2$$

$$s_{pool}^2 = \frac{(n_1 - 1)s_1^2 + (n_2 - 1)s_2^2}{n_1 + n_2 - 2}$$

$$\mathbb{E}(s_1^2) = \sigma_1^2$$

$$\mathbb{E}(s_2^2) = \sigma_2^2$$

$$\mathbb{E}(s_{pool}^2) = \sigma^2$$

$$df_1 = n_1 - 1, df_2 = n_2 - 1, df_{pool} = n_1 + n_2 - 2$$

$$\text{MSE} = \hat{\sigma}^2$$